

I. IDENTIFICATION DATA

Thesis title:	Exploring Regret Minimization and Its Variants in Normal-Form Games
Author's name:	Rozálie Bílková
Type of thesis :	bachelor
Faculty/Institute:	Faculty of Electrical Engineering (FEE)
Department:	Department of Cybernetics
Thesis reviewer:	Doc. Ing. Tomáš Kroupa, Ph.D.
Reviewer's department:	Department of Computer Science

II. EVALUATION OF INDIVIDUAL CRITERIA

Assignment	ordinarily challenging
<i>How demanding was the assigned project?</i>	
The assignment had two main goals (the summary of existing approaches to regret minimization and the implementation). The difficulty is on the standard level required at FEL ČVUT.	

Fulfilment of assignment	fulfilled
<i>How well does the thesis fulfil the assigned task? Have the primary goals been achieved? Which assigned tasks have been incompletely covered, and which parts of the thesis are overextended? Justify your answer.</i>	
The assignment was fulfilled to a good degree. The author provides a theoretical overview of the relevant game-theoretic background and regret-minimization algorithms, implements GRM, and evaluates it on a broad collection of benchmark games. The experimental section is particularly extensive for a bachelor thesis. It includes both iteration-based comparisons against RM, RM+, PRM, MWU, and Fictitious Play, and wall-clock-time comparisons against Double Oracle.	

Activity and independence when creating final thesis	A - excellent.
<i>Assess whether the student had a positive approach, whether the time limits were met, whether the conception was regularly consulted and whether the student was well prepared for the consultations. Assess the student's ability to work independently.</i>	
Rozálie worked mostly independently and followed my advice regarding the theoretical part and the setting for the experiments. She was able to finish the thesis in the extended time despite the health issues.	

Technical level	B - very good.
<i>Is the thesis technically sound? How well did the student employ expertise in his/her field of study? Does the student explain clearly what he/she has done?</i>	
The thesis demonstrates a solid understanding of the basic theory of finite zero-sum games and no-regret learning. The explanation of the minimax formulation, external regret, MWU, Regret Matching, and GRM is generally appropriate for the level of a bachelor thesis. The author correctly emphasizes exploitability as the main performance measure and uses it consistently throughout the empirical evaluation. Some mathematical statements should be formulated with greater precision and with explicit assumptions. The experimental comparison with Double Oracle is useful but would be stronger with a unified solver methodology or a clearer discussion of how the solver choice affects runtime and convergence.	

Formal level and language level, scope of thesis	C - good.
<i>Are formalisms and notations used properly? Is the thesis organized in a logical way? Is the thesis sufficiently extensive? Is the thesis well-presented? Is the language clear and understandable? Is the English satisfactory?</i>	
The thesis is logically structured and contains all standard parts: introduction, theoretical background, algorithmic description etc. The notation table is helpful, and the implementation section is very detailed for a bachelor thesis. The main weakness of the thesis is the language and formal presentation. The text contains many typographical, grammatical, and spelling errors, for example "acelerate," "dymanics," "desribe," "expecially," "fiminishing," "subliear," "loewr," "mych," and similar errors. These mistakes do not invalidate the technical contribution, but they noticeably reduce readability. There are also minor notational inconsistencies and occasional imprecise formulations, especially in the theoretical sections on correlated equilibrium, Double Oracle, and the relation between payoff and loss formulations.	

Selection of sources, citation correctness**A - excellent.**

Does the thesis make adequate reference to earlier work on the topic? Was the selection of sources adequate? Is the student's original work clearly distinguished from earlier work in the field? Do the bibliographic citations meet the standards?

The references are adequate and include the key paper by Zhang, Lerer, and Brown on Greedy Regret Minimization, as well as standard sources on algorithmic game theory, MWU, Regret Matching, and Double Oracle.

Additional commentary and evaluation (optional)

Comment on the overall quality of the thesis, its novelty and its impact on the field, its strengths and weaknesses, the utility of the solution that is presented, the theoretical/formal level, the student's skillfulness, etc.

III. OVERALL EVALUATION, QUESTIONS FOR THE PRESENTATION AND DEFENSE OF THE THESIS, SUGGESTED GRADE

Summarize your opinion on the thesis and explain your final grading.

The thesis addresses a relevant and current topic in algorithmic game theory and learning in games. It implements a nontrivial recent algorithm and evaluates it broadly. The experimental comparison is ambitious and includes several meaningful baselines. The thesis also identifies limitations of GRM rather than presenting it as universally superior, which makes the conclusions more balanced and scientifically credible. The principal weaknesses are formal rather than conceptual.

Questions:

1. The thesis uses loss functions in the theoretical exposition but payoff functions elsewhere. How exactly are the regret updates and exploitability computations transformed between these two conventions?
2. Why does GRM perform poorly in transitive games? Can this behaviour be explained from the structure of the greedy weight objective or from the dominance structure of the payoff matrix?

The grade that I award for the thesis is **B - very good**.

Date: **25.5.2026**

Signature: